



US 20250277610A1

(19) **United States**

(12) **Patent Application Publication**
LESAGE

(10) **Pub. No.: US 2025/0277610 A1**

(43) **Pub. Date: Sep. 4, 2025**

(54) **REFRIGERATION SYSTEM USING
SEPARATE SUBCOOLING LOOP WITH
HEAT-CARRYING REFRIGERANT**

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(21) Appl. No.: **19/068,461**

(22) Filed: **Mar. 3, 2025**

Related U.S. Application Data

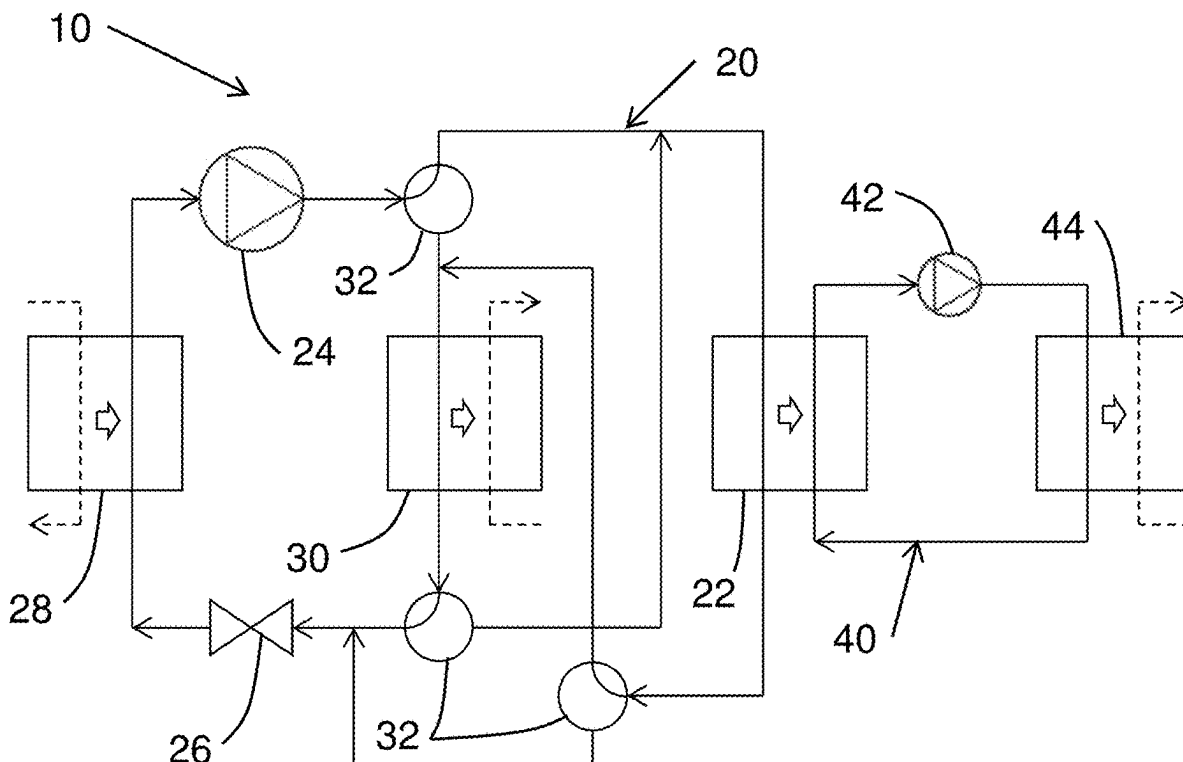
(60) Provisional application No. 63/559,986, filed on Mar.
1, 2024.

Publication Classification

(51) **Int. Cl.**
F25B 40/02 (2006.01)
F25B 6/00 (2006.01)
(52) **U.S. Cl.**
CPC **F25B 40/02** (2013.01); **F25B 6/00**
(2013.01)

(57) **ABSTRACT**

A refrigeration system, having a main loop operating with a first refrigerant and including a first heat exchanger unit connected between a compressor unit and an expansion unit upstream of an evaporation unit, includes a subcooling loop. The subcooling loop has a second refrigerant circulating therein via a pump unit and connects to the first heat exchanger unit to heat-up the second refrigerant. The subcooling loop runs to an air circulation unit located downstream the first heat exchanger unit to cool down the second refrigerant and warm-up air flowing into the air circulation unit.



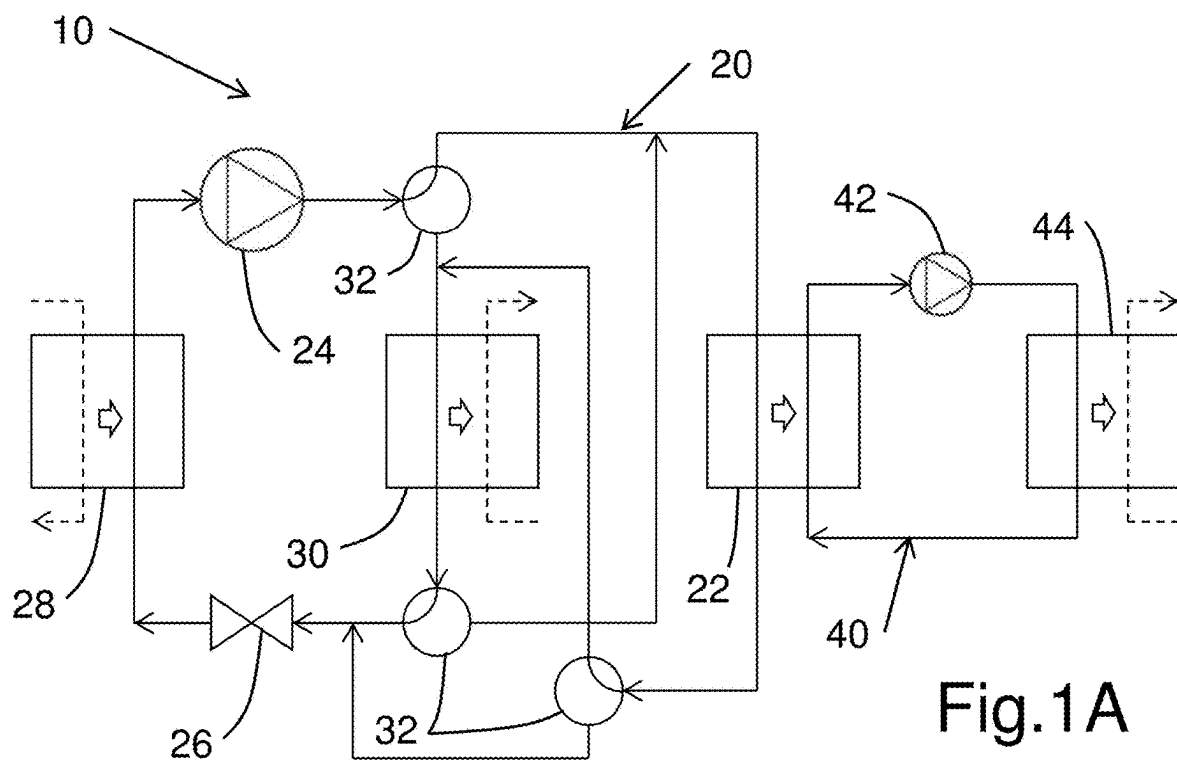


Fig.1A

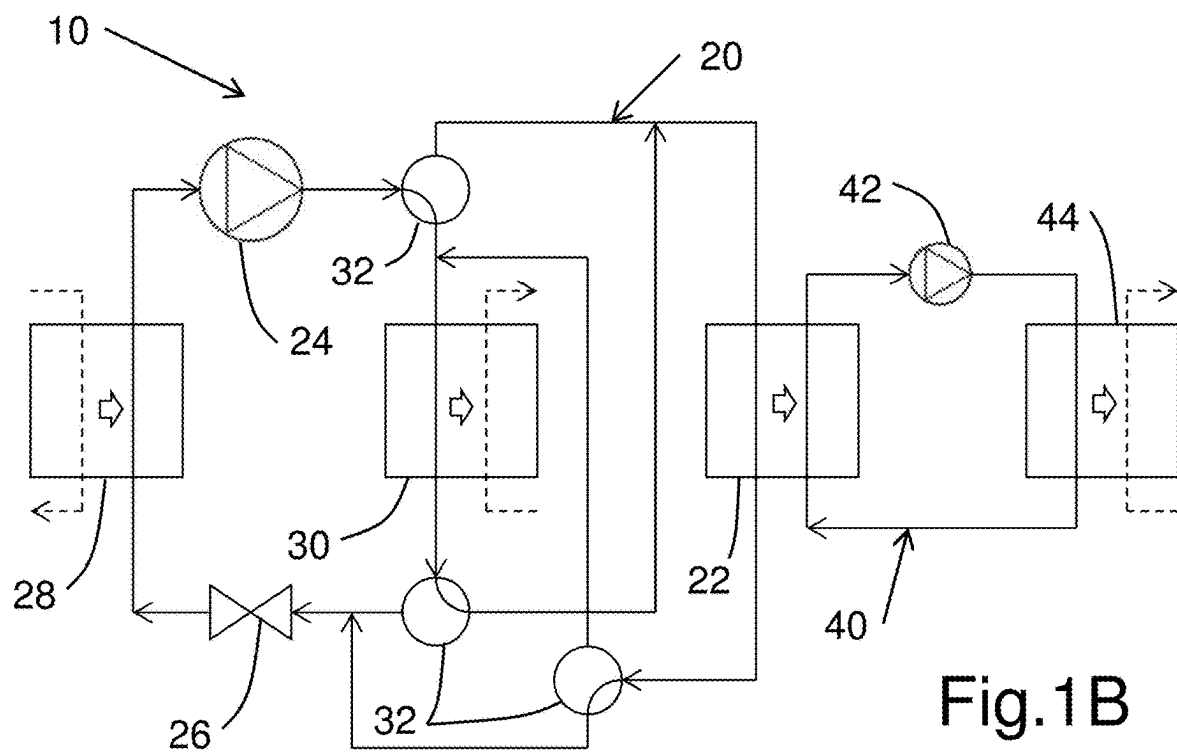
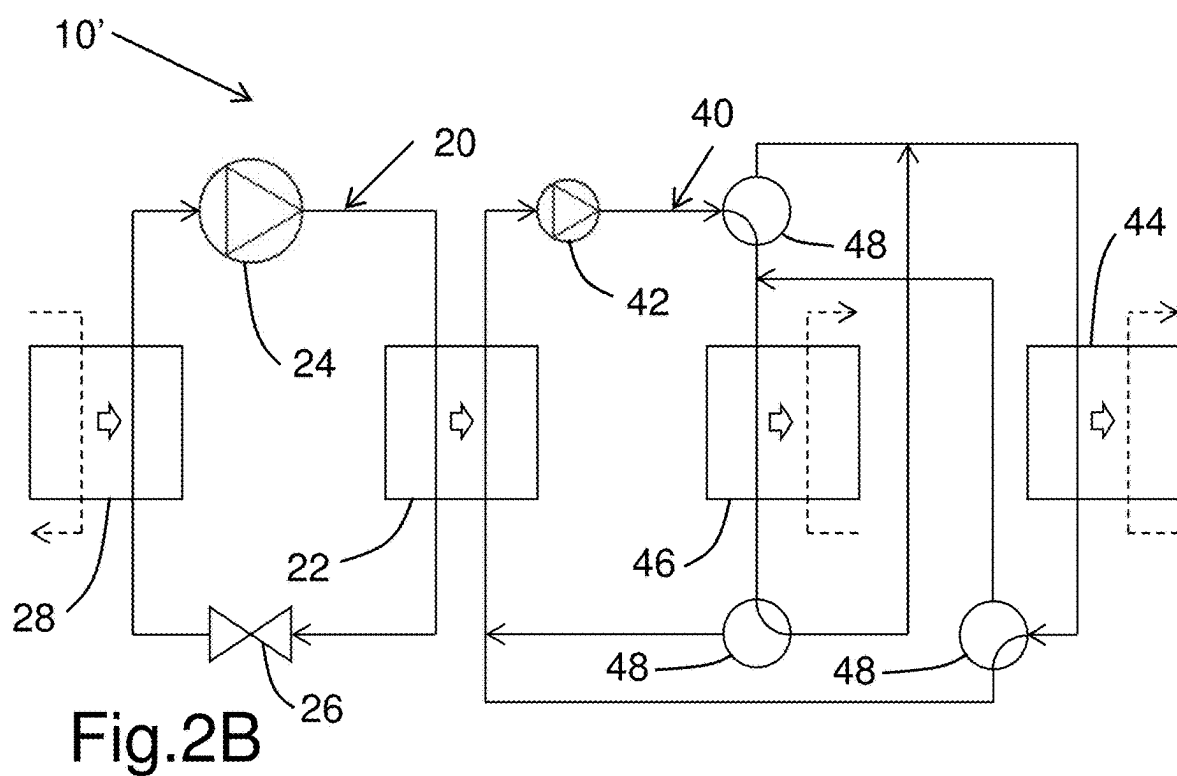
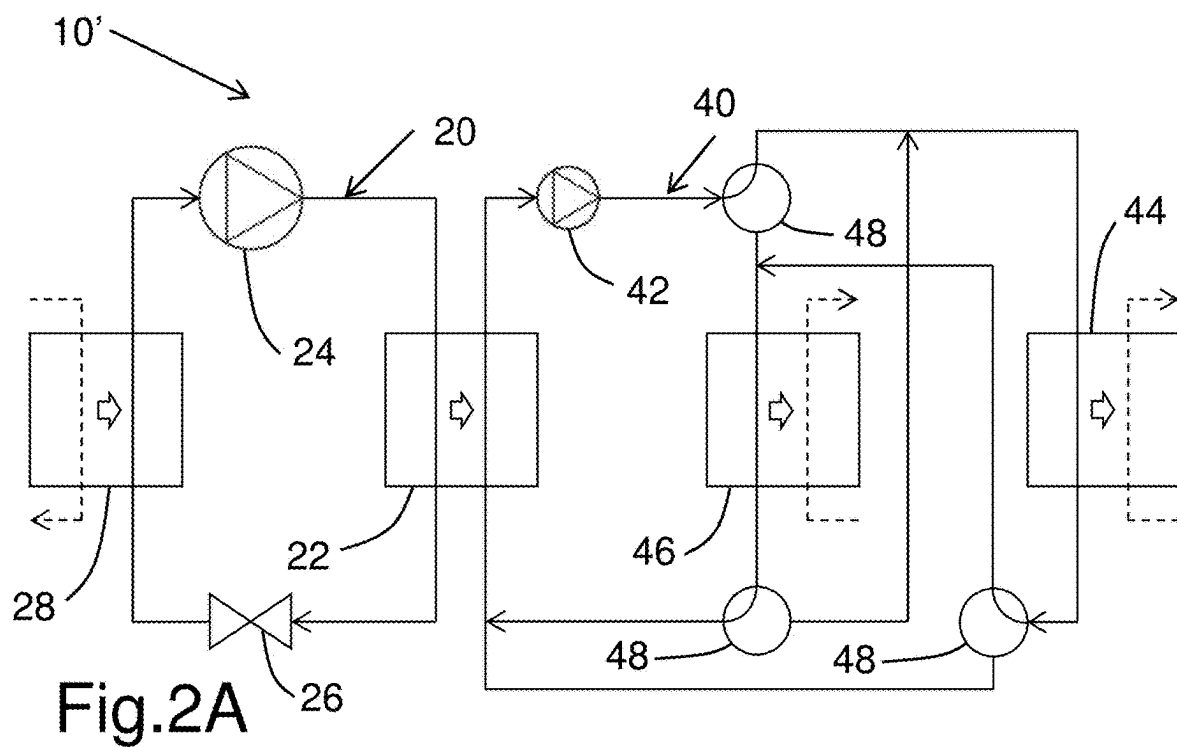


Fig.1B



REFRIGERATION SYSTEM USING SEPARATE SUBCOOLING LOOP WITH HEAT-CARRYING REFRIGERANT

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims the benefit of U.S. provisional patent application No. 63/559,986, filed on Mar. 1, 2024.

TECHNICAL FIELD OF THE INVENTION

[0002] The present disclosure relates to a refrigeration system including heat reclaim. More specifically, the refrigeration system also includes a separate independent subcooling loop with a heat-carrying refrigerant.

BACKGROUND OF THE INVENTION

[0003] Traditionally, in typical refrigeration systems, it is well known to desuperheat the (gaseous) refrigerant coming out of the compressor units to lower the refrigerant temperature down to condensation temperature. Once at the condensation temperature, it is needed to further cool down the (gas and liquid mixture) refrigerant temperature to essentially liquefy all the refrigerant before it reaches the evaporator units downstream of an expansion unit to make the low temperature refrigerant gaseous again. This is typically done using a heat exchanger, such as a condenser or a heat reclaim unit to warm up the air inside a building, especially during chilled or cooled evenings and nights, or during winters.

[0004] To improve the efficiency of the evaporator units, it could be advisable to even further cool down, or subcool the (liquid) refrigerant temperature, and use that latest energy extracted from the refrigerant to heat up another fluid that needs that energy. As the subcooling process is not always possible, since it depends on many parameters such as external conditions, required capacity of the refrigeration system, etc., it is usually rather expensive to build-in the equipment required for such process, especially when using the same main refrigerant.

[0005] Accordingly, there is a need for a refrigeration system having a main refrigeration loop and a separate subcooling loop.

SUMMARY OF THE INVENTION

[0006] It is therefore a general object of the present disclosure to provide a refrigeration system having a main refrigeration loop and a separate subcooling loop that obviates the above-noted drawbacks and problems.

[0007] An advantage of the present invention is that the separate subcooling loop uses a heat-carrying refrigerant, preferably containing a good portion of water (such as glycol water or the like), circulating in a relatively small quantity (because of a small loop, especially when compared to the size of the main loop of the refrigeration system). Accordingly, a preferred usage of the present invention is to use the heat captured by the heat-carrying refrigerant of the subcooling loop is at least partially transferred to an air circulation unit to warm-up the temperature of the air circulating into the unit to essentially lower the humidity level of the air, while maintaining a minimum temperature thereof. This is better performed when warming-up the air

after the air was conditioned, if applicable (depending on the instant/current conditions), when within an air conditioning unit.

[0008] Another advantage of the present invention is that the separate subcooling loop allows for a useful heat recuperation from the main refrigerant to heat-up the heat-carrying refrigerant of the subcooling loop when the air of a room/building needs to be dehumidified, such as in summer periods, when heat is available (as opposed to periods where significant heat reclaim is required to heat-up the building during winter periods).

[0009] Other advantages of the present invention is that, especially in comparison with a conventional refrigeration system using desuperheating in dehumidification mode, the separate subcooling loop allows the following points.

[0010] The ‘cold or frigorific production’ of the evaporator units of the main refrigeration loop is increased. And this implies that the work to be performed by the compressor units is reduced (as the pressure levels are reduced), with a reduced compressing ratio, a more ‘normal’ (not at the extreme) mechanical working effort from the compressor units, and an improved lubrication thereof. This further helps increasing the life time of the compressor units while reducing the quantity and complexity of maintenance required.

[0011] A reduction of the required supplied energy to the compressor units.

[0012] A reduction of the amount of energy to be released via the condensing unit (especially a fluid cooler or the like) to help increasing the life time thereof.

[0013] According to an aspect of the present disclosure there is provided a refrigeration system having a main loop operating with a first (main) refrigerant and including a first heat exchanger unit connected between a compressor unit and an expansion unit upstream of an evaporation unit, the system comprising:

[0014] a (separate) subcooling loop having a second (heat-carrying) refrigerant circulating therein using a pump unit and connecting to the first heat exchanger unit to heat-up the second refrigerant, the subcooling loop running to an air circulation unit located downstream the first heat exchanger unit to cool down the second refrigerant and warm-up air flowing into the air circulation unit.

[0015] In one embodiment, the main loop further including a main heat exchanger (or air condenser) unit and valve units located between the compressor unit and the expansion unit, the valve units selectively connecting the main heat exchanger unit downstream of the first heat exchanger unit to desuperheat the first refrigerant before being cooled down using the main heat exchanger unit, the valve units further allowing to selectively connect the main heat exchanger unit upstream of the first heat exchanger unit to enable the first heat exchanger unit subcooling down the first refrigerant.

[0016] In one embodiment, the subcooling loop further including a second heat exchanger (or air condenser) unit and valve units, the valve units selectively connecting the second heat exchanger unit downstream of the air circulation unit and upstream of the first heat exchanger to desuperheat the second refrigerant before being cooled down using the second heat exchanger unit, the valve units further allowing to selectively connect the second heat exchanger unit upstream of the air circulation unit to enable the air circu-

lation unit further cooling down the second refrigerant before reaching the first heat exchanger unit to subcool the first refrigerant.

[0017] Other objects and advantages of the present invention will become apparent from a careful reading of the detailed description provided herein, with appropriate reference to the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] Embodiments of the disclosure will be described by way of examples only with reference to the accompanying Figures, with similar references referring to similar components, in which:

[0019] FIGS. 1A and 1B are simplified schematic diagrams of a refrigeration system in accordance with an embodiment of the present invention, showing a subcooling loop having a heat-carrying refrigerant selectively connected to a main loop, with the subcooling loop operating in a desuperheating mode and subcooling mode of the main refrigerant, respectively; and

[0020] FIGS. 2A and 2B are simplified schematic diagrams of a refrigeration system in accordance with another embodiment of the present invention, showing a subcooling loop having a heat-carrying refrigerant selectively connected to a main loop, with the subcooling loop operating in a desuperheating mode and subcooling mode of the heat-carrying refrigerant, respectively, while cooling down the main refrigerant.

DETAILED DESCRIPTION OF THE INVENTION

[0021] Referring to FIGS. 1A-2B, there is shown an embodiment of a refrigeration system 10, 10' in accordance with the present invention, as typically used in a supermarket, grocery store or the like, and for which the main refrigerant loop/circuit is relatively large and becomes expensive when maintenances or repairs are needed. The refrigeration system 10, 10' has a main loop 20 operating with a first (or main) refrigerant and typically includes a first heat exchanger unit 22 connected between a compressor unit 24 and an expansion unit 26 upstream of an evaporation unit 28 (to cool down a fluid or preferably air circulating therethrough, as represented by the stippled line path). The first heat exchanger unit 22 typically reduces the temperature of the main refrigerant to condensate the same.

[0022] The system 10, 10' further includes a separate (independent) subcooling loop 40 having a second (or heat-carrying) refrigerant circulating therein via a pump unit 42. The subcooling loop 40 is operatively connected to the main loop 20 at the first heat exchanger unit 22 to transfer heat from the main refrigerant to the heat-carrying refrigerant. The subcooling loop 40 typically runs to an air circulation unit 44 located downstream the first heat exchanger unit 22 to transfer at least a portion of the heat of the heat-carrying refrigerant to the air flowing through the air circulation unit 44 (as represented by the stippled line path), in order to cool down the heat-carrying refrigerant and warm-up the air so as to dehumidify the air that is ultimately circulated to a room/building (not shown). The air circulation unit 44 is typically part of an air conditioning system (not shown) that conditions the air of the room/building whenever required.

[0023] The first heat exchanger 22 is typically a plate heat exchanger, or a shell and tube heat exchanger or a coil heat exchanger or the like. The heat-carrying refrigerant used in the subcooling loop 40 is preferably a water-based refrigerant, such as glycol water or the like, especially to lower maintenance cost and complexity.

[0024] Referring now more specifically to FIGS. 1A and 1B, in this embodiment 10, the main loop 20 further includes a main heat exchanger unit 30 (such a heat reclaim unit, or an air condenser unit, or a liquid-cooled condenser or the like) to transfer heat from the main refrigerant to another fluid (such as outdoor or ambient air, or another third refrigerant, as represented by the stippled line path) also circulating therethrough, and valve units 32 located between the compressor unit 24 and the expansion unit 26. The valve units 32, such as an arrangement of three-way valves (as illustrated in the Figures) or the like, selectively connect the main heat exchanger unit 30 downstream of the first heat exchanger unit 22 to desuperheat the main refrigerant before it gets cooled down using the main heat exchanger unit 30, as illustrated in FIG. 1A.

[0025] Alternatively, the valve units 32 further allow to selectively connect the main heat exchanger unit 30 upstream of the first heat exchanger unit 22 to enable the first heat exchanger unit 22 to subcool down the main refrigerant, as illustrated in FIG. 1B, when only a relatively small amount of heat is required to simply dehumidify the air circulating through the air circulation unit 44, such as in summertime when heat recuperation via the main heat exchanger unit 30 or the like is not in high demand (which would be the case in winter time and the like).

[0026] Referring now more specifically to FIGS. 2A and 2B, in this embodiment 10', the subcooling loop 40 further includes a second heat exchanger unit 46 (such a heat reclaim unit, or an air condenser unit, or a liquid-cooled condenser or the like, as represented by the stippled line path) to transfer heat from the heat-carrying refrigerant to another fluid (such as outdoor or ambient air, or another third refrigerant) also circulating therethrough, and valve units 48 located between the first heat exchanger unit 22 and the air circulation unit 44. The valve units 48, such as an arrangement of three-way valves (as illustrated in the Figures) or the like, selectively connect the second heat exchanger unit 46 downstream of the air circulation unit 44 and upstream of the first heat exchanger 22 to desuperheat the second refrigerant before it gets cooled down using the second heat exchanger unit 46, as illustrated in FIG. 2A.

[0027] The valve units 48 further allow to selectively connect the second heat exchanger unit 46 upstream of the air circulation unit 44 to enable the air circulation unit 44 to further cool down the second refrigerant before it reaches the first heat exchanger unit 22 and subcool the first refrigerant, as illustrated in FIG. 2B, when only a relatively small amount of heat is required to simply dehumidify the air circulating through the air circulation unit 44, such as in summertime when heat recuperation via the second heat exchanger unit 46 is not in high demand (which would be the case in winter time and the like).

[0028] Although the embodiment 10, 10' illustrated in the Figures schematically show only the required components to illustrate the present invention, one skilled in the art would readily understand that any other additional components as other heat exchanger units, valve units, additional refrigerant

loops/subloops and the like could be considered without departing from the scope of the present invention.

[0029] Although the present disclosure has been described with a certain degree of particularity and by way of an illustrative embodiment and examples thereof, it is to be understood that the present disclosure is not limited to the features of the embodiments described and illustrated herein, but includes all variations and modifications within the scope and spirit of the disclosure as hereinafter claimed.

1. A refrigeration system having a main loop operating with a first refrigerant and including a first heat exchanger unit connected between a compressor unit and an expansion unit upstream of an evaporation unit, the system comprising:

a subcooling loop having a second refrigerant circulating therein using a pump unit and connecting to the first heat exchanger unit to heat-up the second refrigerant, the subcooling loop running to an air circulation unit located downstream the first heat exchanger unit to cool down the second refrigerant and warm-up air flowing into the air circulation unit.

2. The system of claim 1, wherein the main loop further including a main heat exchanger unit and valve units located

between the compressor unit and the expansion unit, the valve units selectively connecting the main heat exchanger unit downstream of the first heat exchanger unit to desuperheat the first refrigerant before being cooled down using the main heat exchanger unit, the valve units further allowing to selectively connect the main heat exchanger unit upstream of the first heat exchanger unit to enable the first heat exchanger unit subcooling down the first refrigerant.

3. The system of claim 1, wherein the subcooling loop further including a second heat exchanger unit and valve units, the valve units selectively connecting the second heat exchanger unit downstream of the air circulation unit and upstream of the first heat exchanger to desuperheat the second refrigerant before being cooled down using the second heat exchanger unit, the valve units further allowing to selectively connect the second heat exchanger unit upstream of the air circulation unit to enable the air circulation unit further cooling down the second refrigerant before reaching the first heat exchanger unit to subcool the first refrigerant.

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